

Module 3.5 — Practice Day 01

Book Library Management System

Topics: GET • POST • Path Parameters • Query Parameters

Overview:

You will build a **Book Library API** from scratch using FastAPI. No database required. Load initial data from a `books.json` file.

Data Model:

Each book has the following fields:

Field	Type	Description
book_id	int	Sent by the client in the POST request body
title	str	Title of the book
author	str	Author name
genre	str	One of: "fiction", "science", "history", "biography"
pages	int	Total number of pages
rating	float	Book rating (e.g. 4.5)

books.json File:

You must create a file named `books.json` in your project folder. When the app starts, read this file and load the books into your in-memory list. This means your GET endpoints work immediately without needing to POST first.

books.json content:

```
[
  {
    "book_id": 1,
    "title": "Sapiens",
    "author": "Yuval Noah Harari",
    "genre": "history",
    "pages": 443,
    "rating": 4.8
  },
  {
    "book_id": 2,
    "title": "The Alchemist",
    "author": "Paulo Coelho",
    "genre": "fiction",
    "pages": 208,
```

```
        "rating": 4.5
    },
    {
        "book_id": 3,
        "title": "A Brief History of Time",
        "author": "Stephen Hawking",
        "genre": "science",
        "pages": 212,
        "rating": 4.6
    }
]
```

How to load it in main.py:

```
import json

with open('books.json', 'r') as f:
    books = json.load(f)
```

Required Endpoints:

1. GET /books

Return all books currently in the books.json file.

2. GET /books/{book_id}

Return a single book matching the given `book_id`.

- **If found:** return the book object with status 200.
- **If not found:** return a meaningful error message with status 404. Do not crash.

3. POST /create_books

Add a new book to the library.

- The client sends all fields in the request body, including `book_id`.
- Append the new book to the in-memory list.
- Return the newly created book object with status 201.

4. GET /books/sort

Return all books sorted by a field, in a given direction.

Query Parameters:

Parameter	Allowed Values	Default
sort_by	pages, rating	rating
order	asc, desc	desc

Example: `/sort?sort_by=pages&order=asc`